2024 3rd International Conference for Innovation in Technology (INOCON)
Karnataka, India. Mar 1-3, 2024

2024 3rd International Conference for Innovation in Technology (INOCON) | 979-8-3503-8193-1/24/$31.00 ©2024 IEEE | DOI: 10.1109/INOCON60754.2024.10511976

Cognitive Empowerment: Dyslexic Reading and Learning Support System

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***Abstract*—In the current digital era, this research project strives to empower individuals with dyslexia through the development of an innovative software solution that addresses their distinct challenges in handwritten document conversion, language error correction, and language skill enhancement. The proposed system integrates advanced Optical Character Recognition (OCR) technology, natural language processing algorithms, personalized practice sessions, and voice guidance to precisely convert handwritten text, identify and rectify spelling, grammar, and punctuation errors, and provide customized opportunities for language improvement. With a focus on accessibility and user-friendliness, the project bridges the divide between dyslexic individuals and effective communication, learning, and documentation in the digital domain, representing a significant advancement in inclusivity and support for this community. The conversion of handwritten documents to digital text and subsequent correction utilizes a grammatical correction model implemented in Python. The obtained results are thoroughly examined, and the paper discusses the system's future prospects.**

Keywords— Dyslexia, Optical Character Recognition (OCR), language correction, language enhancement, accessibility, user-friendly design, digital communication, learning support.

I. Introduction

Dyslexia, a neurodevelopmental disorder, poses significant challenges to individuals in mastering reading, writing, and spelling skills. Rooted in difficulties with phonological processing, dyslexia often manifest in distinctive handwriting characteristics, such as irregular letter shapes, inconsistent spacing, and reversed or transposed letters or words.

Current OCR and text correction systems face inherent limitations when dealing with dyslexic handwriting. The irregularities present in such writing styles often result in inaccuracies, making it difficult for standard algorithms to recognize and correct the text effectively.[2]

Conventional detection of dyslexia often focuses on behavioral aspects, including reading and writing proficiency, IQ, phonological awareness (sounding out words), and working memory, typically all assessed using standardized tests. This process is time consuming, and can miss many cases of dyslexia since every person experiences it differently.[1]

In response to these challenges, our project sets out to develop a transformative software solution. Beyond the conventional conversion of handwritten documents into digital text, our approach incorporates a dyslexic detection and correction model. This model integrates advanced Optical Character Recognition (OCR) technology, leveraging natural language processing algorithms. Moreover, it incorporates personalized practice sessions and voice guidance to ensure accurate identification and correction of spelling, grammar, and punctuation errors unique to dyslexic handwriting.[3]

Implemented using Python, our system places a strong emphasis on accessibility and user-friendliness. The goal is to narrow the divide between dyslexic individuals and effective communication, learning, and documentation in the digital domain. The envisioned software aims to benefit individuals with dyslexia by providing them with a tool that goes beyond standard OCR and correction systems. By offering accurate, personalized, and supportive correction, our system has the potential to empower individuals with dyslexia to communicate more effectively and confidently in various digital contexts.

II. Technology Stack/Dyslexic Detection And Training

A. Dyslexic Handwritten Text Recognition OCR model

The Dyslexic Handwritten Text Recognition OCR model is a Convolutional Neural Network (CNN) built in Keras using the Sequential model, with a specific focus on addressing dyslexic handwriting challenges. Three convolutional layers are followed by max-pooling layers for spatial reduction, dropout layers for regularization, and dense layers for classification in the architecture. The model is tailored for recognizing 79 different classes, likely representing characters or symbols related to dyslexic handwriting, with 79 neurons in the output layer and SoftMax activation. The training dataset contains 299,211 examples, and the testing dataset contains 18,989 examples. This large dataset, which includes both

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uppercase and lowercase letters, provides the model with a diverse training environment.

One distinguishing feature of this model is the use of inverted images to improve its ability to detect dyslexic text. To accomplish this, the original image data, which has been resized to 32x32 pixels, is supplemented with horizontally flipped versions. This augmentation technique improves the model's generalization to different dyslexic handwriting styles by capturing both the regular and inverted forms of characters. The model's binary output design is especially suited to dyslexic text recognition, with a regular 'C' denoted as 'C' and an inverted 'C' denoted as '(C)'. The model's utility extends beyond its training and evaluation phases; it stores the OCR model using the Pickle library. This allows for easy access during predictions, making the deployment of the dyslexic text detection model in real-world applications more efficient.

B. Mistakes Detection

The Mistakes Detection module makes use of Python libraries such as 'language-tool-python' for grammatical checks and 'pyspellchecker' for spelling corrections to ensure comprehensive error detection from the letter to the sentence level. This module has two functions: 'detect_grammar_mistakes' uses 'language-tool-python' to identify grammatical errors, and 'detect_spelling_mistakes' uses 'pyspellchecker' to detect spelling mistakes. The main function demonstrates the module's functionality with an example text that contains mistakes on purpose, and prints the identified grammatical and spelling errors. This strong approach allows for thorough and precise error detection, improving text quality and accuracy.

C. Mistakes Correction

By storing detected errors in a MongoDB database for each user, the Correction module seamlessly integrates with the Mistakes Detection module. Using the SpellChecker library, it meticulously corrects spelling errors using the 'correct_spelling' function. The module identifies and stores detected errors in the database based on user input, ensuring a personalized and efficient tracking system. It then provides the user with a refined text that is free of spelling mistakes, resulting in a more streamlined experience. This method not only improves the overall correction process, but it also lays the groundwork for continuous improvement and tailored error correction in subsequent interactions.

D. Text-to-Speech

By providing audio renditions of webpage content and corrections made in the Correction module, the Text-to-Speech module improves user accessibility. The module converts text inputs into clear and articulate speech using gTTS (Google Text-to-Speech). This feature is designed specifically for dyslexic users, making it easier to absorb information, improve understanding, and facilitate a more seamless learning experience. The module aims to improve accessibility by audibly presenting webpage content and correction details, making information more understandable and assisting dyslexic users in their educational journey. The gTTS library converts text to speech efficiently, enhancing the user experience with auditory learning support.

E. Practice Session

The Practice Session module uses MongoDB-stored user mistakes from the Correction module to generate engaging tracing and spelling practice sets dynamically. The module, created with HTML, CSS, and JavaScript, ensures an appealing and user-friendly interface, enhancing the learning experience for dyslexic users. Tracing exercises are presented using interactive HTML canvases, which allow users to fine-tune their letter formations. At the same time, gTTS is used for spelling practice, providing audio cues. The lessons have been thoughtfully designed with an entertaining user interface to foster a positive and enjoyable learning environment while addressing dyslexic challenges. This integration improves user engagement while also promoting effective learning via personalized and interactive practice sessions.

III. DYSLEXIC DETECTIONSYSTEMS

A. Existing System

Systems that are now in place for learning help and dyslexic reading prediction include a wide range of technologies that are designed to give dyslexic people individualized support. Text-to-Speech (TTS) software converts written text into spoken speech, making reading easier and promoting aural understanding. Examples of TTS software include Kurzweil 3000 and NaturalReader. For dyslexic people, specialized typefaces like Dyslexie and OpenDyslexic, as well as background colour options that may be customized, are intended to lessen visual stress and improve reading. Screen readers that read aloud on-screen text, like JAWS and NVDA, provide an aural option for people who might find traditional visual reading difficult. Spelling problems can be helped by word prediction programs like WordQ and Co: Writer, which provide word suggestions as users write. Software for speech recognition, such as Dragon NaturallySpeaking, enables computer operation without using hands and spoken commands to text input, providing another way to engage. Nessy Reading and Spelling, among other educational applications and games, use technology to create fun, interactive learning environments that are especially meant to help dyslexic people become better readers. Together, these tools address several aspects of dyslexic reading and learning issues and create a more welcoming and encouraging learning environment.

B. Issues faced with existing system

Existing System for dyslexic mostly only detects whether the person is dyslexic or not and doesn’t provide correction and proper practice sessions. Here are some of the issues with manual student attendance systems:

1) Manual Correction Dependency:

Spelling, grammar, and punctuation mistakes in handwritten material are frequently corrected by dyslexic people using manual correction techniques. Manual correction can be tedious and time-consuming. It might not always result in extremely accurate adjustments, which presents a serious problem for dyslexic people who want precise and effective remedies.

2) Accessibility Features:

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Tools and software already in use might not have accessibility features designed with dyslexic users' requirements in mind. It might be challenging for dyslexic people to use digital tools for language training since they have trouble using normal user interfaces and functionalities. Their processes of learning and correction may be hampered by this lack of adaptability.

3) Incompatibility with Dyslexic Handwriting:

The distinctive features of dyslexic handwriting may not be adequately handled by many OCR and handwriting recognition programmes since they are made for conventional handwriting. Handwriting characteristics that are characteristic of dyslexics include irregular spacing, inconsistent letter forms, and fluctuating letter sizes.[2] These changes might not be taken into consideration by current systems, leading to erroneous conversions.

4) Pronunciation Improvement Support:

One of the biggest challenges for many dyslexic people is improving their pronunciation, which may not receive enough support from the language learning resources now available. The capacity to appropriately enunciate words can be affected by dyslexia.[3] Language learning aids may not adequately address this problem without specialist assistance, which would impede dyslexic learners' ability to pronounce words clearly and accurately.

5) Feedback and Progress Tracking:

Current methods might not be able to provide thorough feedback on writing mistakes or monitor language learning progress over time. Like any learners, dyslexics need regular, helpful feedback to advance their language abilities. They find it difficult to track their progress and monitor their development in the absence of such feedback.

IV. Beneficiaries

A. Empowering Individuals with Dyslexia:

The dyslexic detection and correction project stands as a beacon of empowerment for individuals grappling with dyslexia. By providing accurate text corrections and voice guidance for pronunciation, the project enables effective communication and enhances writing skills. The personalized practice sessions contribute to increased confidence in academic and professional settings. Moreover, the integration of dyslexic handwriting detection ensures accessibility, making the transformative capabilities of the software readily available to those who need them most.

B. Fostering Inclusive Learning Environments:

Educational institutions benefit from the project's commitment to inclusivity. It addresses the unique challenges faced by dyslexic students, creating a more accessible and supportive learning environment. The correction model's ability to track users' progress allows educators to tailor interventions based on individual needs, ultimately enhancing the overall learning experience and ensuring that no student is left behind.

C. Driving Innovation in Technology:

For technology and software developers, the project provides valuable insights into addressing challenges related to dyslexia. By contributing to the advancement of technology in inclusive software design, developers can create innovative solutions that cater to a broader spectrum of users. The correction model and dyslexic detection features can be further integrated into various assistive technology applications, expanding the reach of inclusive technology solutions.

V. Flow Of Modules

The proposed model is a Python application used to convert the handwritten image to digital text, detect whether dyslexic or not, does all the necessary corrections and provide required practice sessions to the users.

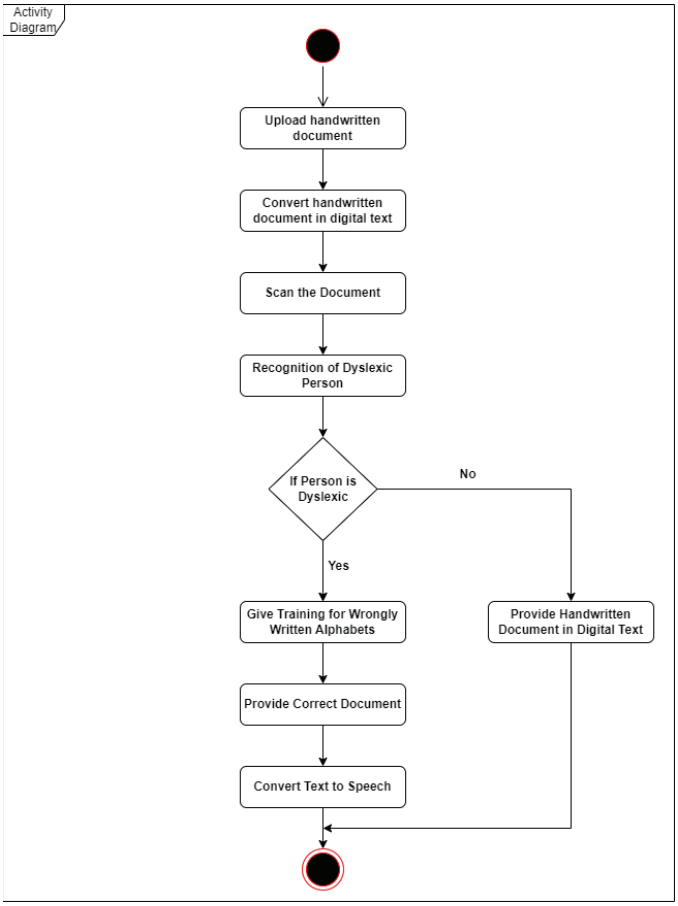


Fig. 1. Activity Diagram

1) OCR Module: During this module, the input which is expected is image of handwritten text: the image is passed through the OCR module where the handwritten text is converted to digital text.

2) Detection Module: This module takes the scanned digital text as input and detects from the text whether it is dyslexic or not.

3) Correction Module: This model is designed to detect and rectify various types of errors, including spelling, grammar, punctuation, and writing idiosyncrasies, specific to

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dyslexic handwriting. Also, the errors and corrections are stored in database for further modules.[3]

4) Text to Speech Module: This module receives the corrected text from the previous modules. It converts written text into spoken language, which can greatly benefit those who may have difficulty reading or comprehending written content.

5) Practice Session Module: This module leverages the mistakes stored in the database and offers two types of practice sessions: letter tracing and word pronunciation. For the required practice session, the module retrieves relevant mistakes and errors from the database.

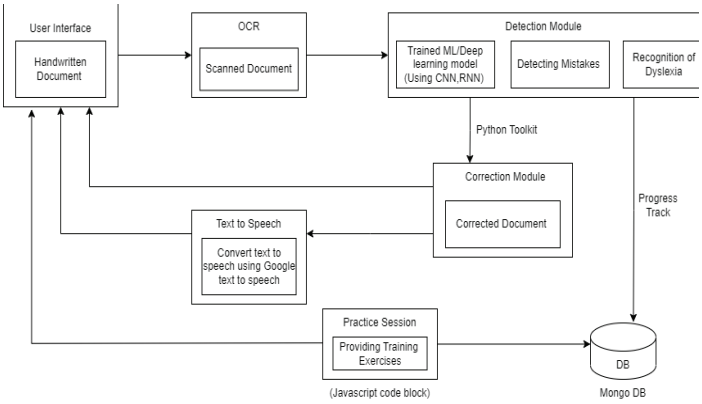


Fig. 2. Flow Of Modules

The above diagram displays an outline for the flow of modules implemented in our project.

VI. Result Analysis

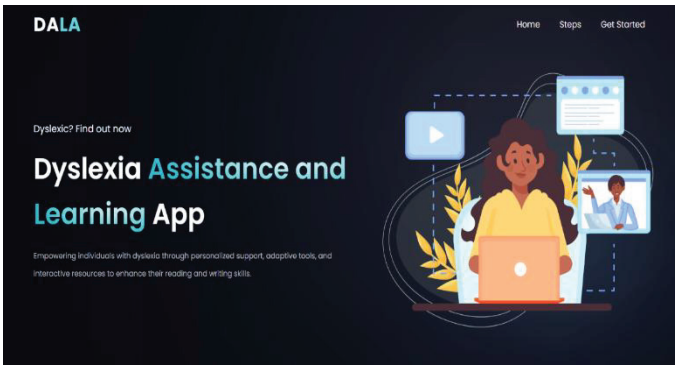


Fig. 3. Home Page

This is our website main page to provide dyslexic reading and learning assistance.

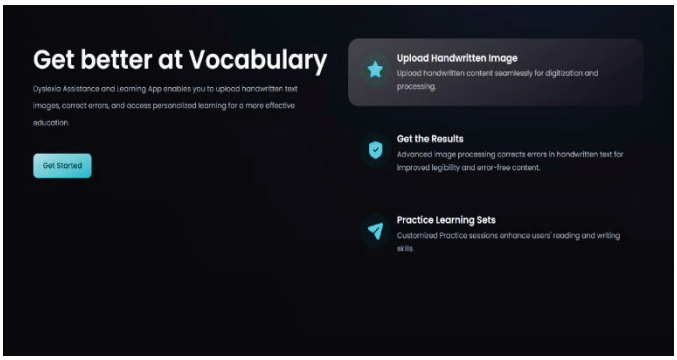


Fig. 4. Steps in Project

Hereweareshowing the various steps that are performed in our project.

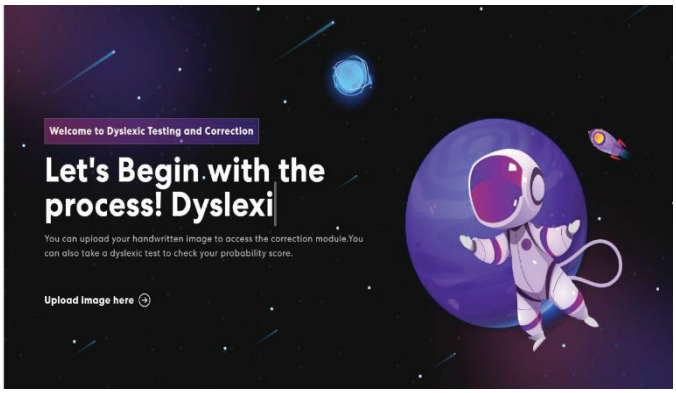


Fig. 5. Upload image

Here we are allowing the users to upload a handwritten image in order to detect whether the person is dyslexic or not and to access the various other modules.

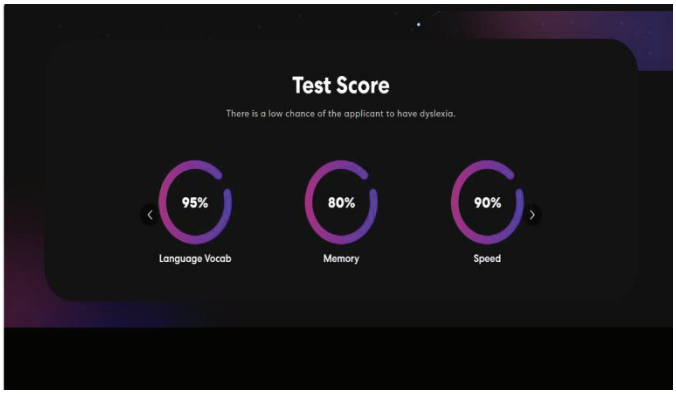


Fig. 6. Test Score

Here we are providing the users with various test such as language vocab, memory test etc to detect whether the person is dyslexic or not.

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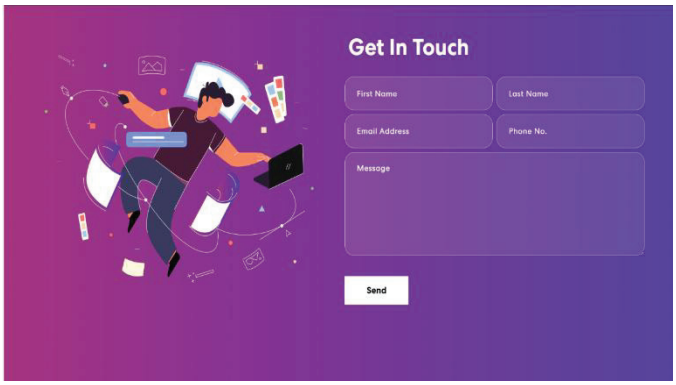


Fig. 7. Contact Us Page

This is our website contact us page to get in touch with us for any further queries or feedback.

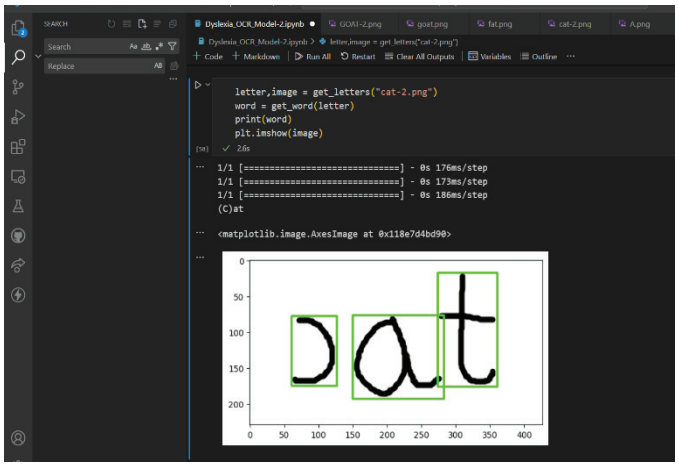


Fig. 8. Dyslexic Detection Model

This is the dyslexic detection module that tracks the various inverse letters and keep count of it to identify whether a person is dyslexic or not based on the given threshold.

VII. Future Scope

In envisioning the future scope of our project, our primary goal is to ensure a continuous evolution that effectively meets the dynamic and evolving needs of dyslexic individuals. To achieve this, we plan to enhance the software's capabilities significantly by incorporating advanced machine learning algorithms. This integration will enable the system to adapt and learn from user interactions over time, fostering an adaptive learning approach. The adaptive learning mechanism aims to refine the software's ability to identify and correct errors, resulting in a more personalized and accurate experience for each user.

Another crucial aspect of our future development involves the incorporation of language models designed to recognize and accommodate various linguistic nuances. This expansion will not only refine the accuracy of language correction but also broaden the software's applicability across diverse linguistic backgrounds, ensuring inclusivity on a global scale.

Recognizing the importance of accessibility, we are committed to creating a mobile application for our system. This mobile application will extend the reach of our solution, allowing dyslexic individuals to benefit from its transformative capabilities on various devices. The mobile application aims to provide on-the-go support, ensuring that users can access personalized practice sessions, corrections, and voice guidance whenever and wherever they need it.

Furthermore, we plan to explore strategic collaborations with educational institutions and assistive technology providers. This collaborative effort aims to seamlessly integrate our solution into existing learning environments and broader assistive technology ecosystems. By fostering partnerships with key stakeholders, we aim to amplify the positive impact of our project, ensuring that it becomes an integral part of the support system for dyslexic individuals.

In embracing these future developments, our project aspires to continually push the boundaries of inclusivity and support for dyslexic individuals. Through advancements in machine learning, linguistic adaptation, mobile accessibility, and strategic collaborations, we seek to foster a more accessible and empowering learning experience for individuals with dyslexia.

VIII. Conclusion

In conclusion, the dyslexic detection and correction project mark a crucial step towards fostering inclusivity and empowerment for individuals grappling with dyslexia. Going beyond conventional systems, our project employs cutting-edge technologies such as advanced Optical Character Recognition (OCR), natural language processing (NLP) algorithms, and voice guidance to specifically address the distinctive challenges presented by dyslexic handwriting.

Our approach involves not only converting handwritten documents into digital text but also utilizing NLP to decipher the meaning and context of the text. This goes beyond simple recognition, ensuring that the software comprehends the nuances and intent of the user's writing. The commitment to accuracy is underscored by our correction model, which meticulously checks for errors in spelling, grammar, and dyslexia-specific writing patterns.

The project doesn't stop at correction but extends to personalized practice sessions. If the software identifies a wrongly written word, it initiates a training section tailored to the user's needs. This innovative feature ensures that users receive targeted support for areas where improvement is needed, contributing to a continuous learning experience.

Voice assistance is another integral aspect of our project, providing users with a multi-sensory learning experience. This feature aids not only in pronunciation but also in improving language skills. Users can listen to correct pronunciations, enhancing their auditory learning and communication abilities.

In essence, our project is not just a correction tool; it's a comprehensive system that encompasses understanding, correction, training, and auditory support. The commitment to accessibility ensures that individuals with dyslexia can effectively leverage these transformative capabilities. By

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enhancing language skills and facilitating effective communication, our project stands as a testament to the positive impact technology can have on addressing learning differences, creating a more inclusive and supportive digital environment.

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